

Computational Neuroscience:

Analysis of brain patterns using machine learning algorithms

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Progress Report

1 Abstract

Perceptual decisions are often assumed to rely exclusively on current sensory information. However, recent research suggests that previous choices can bias how new evidence is interpreted, promoting perceptual confirmation biases and reducing sensitivity to contradictory information. Understanding how these biases emerge is important for explaining how humans maintain or revise perceptual interpretations over time.

In this thesis, we investigated how prior perceptual decisions influence the integration of repeated sensory evidence. Participants repeatedly categorized sequences of oriented visual stimuli, allowing the study of how earlier responses shape later perceptual judgments when the same or highly similar information is presented again. The project specifically examined whether participants simply tend to repeat previous responses or whether earlier decisions actively bias the accumulation of new sensory evidence.

The results revealed a strong tendency to maintain previous perceptual decisions across stimulus repetitions. Importantly, this effect was associated with subjective confidence and could not be explained only by simple response repetition or exact stimulus repetition. Computational evidence accumulation models further showed that models including drift-rate modulations provided a better fit to the data than models based only on starting-point changes, suggesting that previous decisions bias the ongoing accumulation of sensory information.

Altogether, these findings support the idea that perceptual decisions are shaped by previous choices and confidence signals, contributing to the stability of perceptual beliefs over time and providing evidence for confirmation bias mechanisms in perceptual decision-making.

Keywords: perceptual decision-making, confirmation bias, evidence accumulation, drift diffusion model, metacognition, confidence, changes of mind, computational neuroscience.

2 Introduction and state of the art

2.1 Perceptual decision-making and evidence accumulation

Humans constantly make perceptual decisions under uncertain conditions in everyday life. For example, when driving in heavy rain, a person may need to quickly decide whether an object in the distance is a pedestrian or simply a shadow. Similarly, after hearing an unclear sound in a crowded room, we often try to interpret whether someone has called our name or whether the sound came from another conversation. In these situations, the brain must transform noisy and ambiguous sensory information into a coherent decision.

Perceptual decision-making refers to the cognitive process through which individuals interpret sensory information and select one among several possible alternatives. Rather than relying on a single piece of information, the brain continuously combines sensory evidence over time until enough information is available to support a decision. This process can be influenced not only by the quality of the sensory input itself, but also by previous experiences, expectations, attention and internal confidence states.

One of the most influential computational frameworks used to study this process is the Drift Diffusion Model (DDM) (Ratcliff, 1978; Ratcliff & McKoon, 2008). According to this model, sensory evidence is gradually accumulated over time until it reaches a decision threshold associated with one of the possible responses. These models successfully explain both participants' choices and their response times, making them an important tool for understanding human decision-making.

Traditionally, many perceptual decision-making paradigms assumed that each decision was independent from previous ones. However, more recent research has shown that prior choices can bias how new sensory information is interpreted, even when stimuli are independent (Fischer & Whitney, 2014; Urai et al., 2019). These effects, often described as choice history biases or serial dependencies, suggest that perceptual decisions are not completely independent, but are continuously shaped by previous perceptual experiences and responses (Fritsche et al., 2017; Urai et al., 2019).

2.2 Choice history bias and confirmation bias

Several studies have shown that perceptual decisions do not depend only on current sensory evidence, but also on recent response history and earlier choices. This phenomenon, known as choice history bias, describes the tendency of individuals to repeat previous decisions even when the stimuli are independent from each other (Urai et al., 2019). These findings suggest that previous experiences shape how people interpret new sensory information.

Recent research suggests that these biases are not simply caused by a general tendency to repeat responses, but may appear directly during the process of sensory evidence integration. In particular, different studies have shown that information consistent with past responses tends to receive more weight during evidence accumulation, while contradictory information is processed less strongly (Rollwage et al., 2020; Talluri et al., 2018).

Different computational models have tried to explain the origin of these biases. Some authors propose mechanisms based on self-consistent Bayesian inference (Luu & Stocker, 2018), while others suggest that the bias emerges directly during evidence accumulation through selective overweighting of information that is consistent with previous responses (Bronfman et al., 2015; Talluri et al., 2018). In this context, sequential evidence accumulation models are useful for studying different possible bias effects involved in sensory decision-making.

This project is part of this research line and studies how prior decisions modulate the integration of sensory evidence when participants repeatedly process the same sequence of visual stimuli. This paradigm allows us to study how confirmation bias affects the stability and revision of previous perceptual decisions.

2.3 Metacognition, confidence and changes of mind

The ability to review previous decisions is an important aspect of perceptual decision-making. These processes, known as changes of mind, reflect the possibility that new sensory evidence can modify previously made decisions.

The probability of maintaining or changing a decision seems to depend strongly on the level of confidence associated with that response. Perceptual confidence acts as an internal signal about the subjective probability of having made a correct decision and can influence how new information is integrated after an initial choice (Peters et al., 2017; Rollwage et al., 2020). Different studies have shown that decisions made with high confidence tend to be more resistant to revision and promote a biased integration of evidence consistent with previous decisions.

These processes are closely related to metacognition, understood as the ability of individuals to evaluate the accuracy of their own decisions (Fleming & Dolan, 2012). Recent studies suggest that metacognitive efficiency may help individuals detect errors and update decisions when facing contradictory evidence (Rollwage et al., 2020).

In this context, this study investigates how metacognitive factors such as confidence and metacognitive efficiency affect the stability and revision of repeated perceptual decisions.

2.4 Justification of the current project

This thesis studies how previous decisions shape the processing of sensory evidence when participants repeatedly process the same sequence of visual stimuli. Unlike many classical perceptual decision-making paradigms, where each trial is considered independent, this experimental design allows the study of how individuals maintain or revise previous perceptual interpretations when facing partially redundant information.

In this task, participants observe sequences of oriented stimuli and repeatedly classify them as mainly cardinal or diagonal. The controlled repetition of these sequences makes it possible to investigate how strongly initial decisions modulate the later processing of new sensory evidence.

This paradigm is especially useful for studying perceptual confirmation bias. It helps distinguish between a simple tendency to repeat previous responses and a biased processing of sensory evidence consistent with the earlier decision.

To understand these processes, the project combines behavioral analyses, mixed models and computational evidence accumulation models. In later stages, machine learning approaches and electroencephalography (EEG) analyses will also be included to explore the predictive and neural processes involved in visual decision-making.

3 Objectives and hypotheses

3.1 Main objective

The main objective of this project is to study how previous perceptual decisions influence the way people process new sensory information during repeated visual tasks. In particular, this project investigates whether the tendency to maintain previous responses reflects a perceptual confirmation bias, meaning that people tend to interpret new information in a way that supports what they previously decided.

Another important objective is to understand whether confidence influences the stability of perceptual decisions. In other words, the project studies whether people are more likely to maintain a previous decision when they feel more confident about it. Finally, the project also aims to evaluate whether computational models of evidence accumulation can explain the behavioral patterns observed in the experiment.

3.2 Main hypotheses

Hypothesis 1: Previous decisions influence later perceptual judgments.

If previous choices affect later decisions, participants should show a tendency to repeat their earlier responses across stimulus repetitions. This hypothesis would be supported if previous decisions significantly predict later responses in the behavioral analyses. On the other hand, if later responses depend only on the current sensory information, this hypothesis would not be supported.

Hypothesis 2: The observed bias reflects confirmation bias rather than simple response repetition.

If participants show a true confirmation bias, previous decisions should influence how new sensory information is processed. In particular, evidence that is consistent with the previous decision should

receive more weight than contradictory evidence. This hypothesis would be supported if the analyses show asymmetric evidence integration or if drift-rate bias models explain the data better than simple response repetition models.

Hypothesis 3: Confidence increases the probability of maintaining previous decisions.

If confidence acts as a stabilizing signal, participants should be more likely to repeat previous responses when confidence in the earlier decision is high. This hypothesis would be supported if confidence significantly predicts repetition behavior or reduces the probability of changes of mind.

Hypothesis 4: Previous decisions bias the accumulation of new sensory evidence rather than only creating an initial response predisposition.

Different computational models make different predictions about how confirmation bias emerges during decision-making. This hypothesis would be supported if models including drift-rate modulation provide a better fit to the data than models based only on starting-point bias.

Hypothesis 5: Participants' decisions can be predicted from behavioral and computational variables.

If previous choices, sensory evidence and confidence systematically influence behavior, machine learning models should be able to predict whether participants will maintain or change their decisions above chance level.

3.3 Specific objectives

1. To analyze whether participants tend to repeat or change their perceptual decisions across successive repetitions of the same stimulus.
2. To study how sensory evidence, previous decisions and confidence influence participants' behavior.
3. To determine whether the observed bias reflects simple response repetition or asymmetric evidence integration consistent with confirmation bias.
4. To apply Hierarchical Sequential Sampling Models (HSSM) to compare different computational bias mechanisms.
5. To evaluate the reliability of the computational models through simulations and parameter recovery analyses.
6. To implement machine learning techniques to predict participants' decisions from behavioral and computational variables.
7. To analyze, in later stages of the project, the neural correlates of confirmation bias using EEG.
8. To explore the combined predictive value of behavioral and neural variables.

4 Changes with respect to the initial plan

During the development of the project, some aspects of the original plan were refined in order to better adapt the work to the real characteristics of the data and to improve the interpretation of the results. These changes did not modify the main research question, but helped to study the mechanisms involved in perceptual confirmation bias more precisely.

In the first stage, the behavioral analysis was carried out using conventional models. Later, mixed models were incorporated to better represent individual variability. This change allowed more robust estimations that were more consistent with the structure of the experiment.

In addition, the progress of the project showed the need to complement the behavioral analysis with computational accumulation models. For this reason, HSSM were included to compare different computational bias mechanisms related to changes in the starting point and drift rate.

Finally, the development of the project also involved a reorganization of the analysis pipeline, giving priority to behavioral and computational analyses before moving to later stages based on machine learning and EEG analysis. At the same time, the theoretical framework related to confirmation bias, sensory information accumulation and metacognition was further developed, allowing a more coherent formulation of the project objectives.

5 Methodology

5.1 Experimental design

The project is based on a perceptual decision-making task designed to study how previous decisions affect the way new evidence is processed. In this task, participants observed sequences of oriented visual stimuli (Gabor patches) and had to classify them as mainly cardinal or diagonal.

Each sequence consisted of six stimuli presented one after another. Task difficulty was manipulated using a continuous variable called the decision variable (DV), which represented how close the sequence was to cardinal or diagonal orientations. Values close to zero corresponded to ambiguous stimuli, whereas more extreme values represented sequences that were easier to classify because they contained more clearly dominant cardinal or diagonal orientations. In addition, task difficulty was dynamically adjusted using a 1-up/2-down adaptive staircase procedure designed to maintain relatively stable accuracy levels across participants.

The main feature of the paradigm was the controlled repetition of sequences within the same trial. After making an initial decision, participants observed either the exact same stimulus sequence again or a physically different sequence with a similar overall decision variable before giving a second response. This manipulation made it possible to study whether previous decisions biased later perceptual judgments independently of the exact physical repetition of the stimulus.

In addition to categorical decisions, the experiment also recorded response times and subjective confidence ratings indicating how certain participants felt about the correctness of their decisions.

5.2 Dataset and preprocessing

The analyses carried out so far focused on the behavioral data obtained during the perceptual task, including participants' decisions, response times, subjective confidence ratings and variables derived from the experimental design.

Before the statistical and computational analyses, a preprocessing procedure was applied to ensure data quality. Extremely fast or slow response times were considered potential outliers, as they may reflect distractions, lapses of attention or accidental responses rather than normal task performance. To identify these trials, response times were filtered individually for each participant using a criterion based on the participant's own response variability. Specifically, trials with response times exceeding the participant's mean response time plus four standard deviations were removed.

This participant-specific procedure prevented penalizing individuals with naturally slower response speeds while allowing the removal of only extremely unusual responses. Only 63 trials out of 6600 trials (approximately 1% of the dataset) were excluded, suggesting that participants performed the task with a high level of attention and consistency.

5.3 Behavioral analysis

The behavioral analysis aimed to study how previous decisions, sensory evidence and metacognitive variables such as confidence shaped participants' perceptual behavior across stimulus repetitions.

In a first stage, descriptive analyses were carried out to identify general behavioral patterns, including response times, perceptual accuracy, probability of repeating decisions, psychometric curves and the relationship between confidence and changes of mind.

Initially, behavioral effects were explored using participant-level logistic regressions. However, these analyses treated all participants as if they responded similarly to the experimental manipulations, ignoring individual differences in response speed, confidence levels and sensitivity to sensory evidence. To better account for this variability, the analyses were later reformulated using Linear Mixed Models (LMMs) and Generalized Linear Mixed Models (GLMMs).

These models are particularly useful for this type of experiment because they allow the analysis of general behavioral patterns while also taking into account individual differences between participants. In this way, each participant can have their own baseline response tendencies, reducing biased estimations and improving the reliability of the statistical inferences.

Using this approach, the analyses examined how sensory evidence, previous decisions, confidence and recent response history influenced participants' behavior throughout the task. Depending on the analysis, the models focused on variables such as response times, perceptual accuracy, the probability of repeating a decision and changes of mind. As a whole, this approach provided a more reliable characterization of the relationship between previous choices, sensory information and perceptual behavior, and also served as the basis for the later computational modeling analyses.

In addition to subjective confidence, the analyses also included a measure of metacognitive efficiency, which reflects how accurately participants' confidence matched their actual task performance. In other words, this measure evaluates whether participants were able to correctly estimate when their decisions were likely to be correct or incorrect.

To quantify this process, metacognitive efficiency was calculated as the ratio between meta- d' and perceptual sensitivity (d'). While d' measures how well participants discriminated between cardinal and diagonal sequences, meta- d' measures how well confidence ratings distinguish between correct and incorrect responses. Metacognitive efficiency was then computed by dividing meta- d' by d' , making it possible to evaluate metacognitive ability independently of overall perceptual performance.

5.4 Drift Diffusion Modeling with HSSM

Although the behavioral mixed models allowed the identification of statistical relationships between previous decisions, sensory evidence and participants' responses, these analyses alone could not fully explain the cognitive processes underlying the observed biases. In particular, GLMM analyses only considered participants' responses, whereas DDM also incorporated response times, making it possible to study how decisions evolved over time.

To investigate these hidden decisional processes, a computational modeling approach based on Hierarchical Sequential Sampling Models (HSSM) was incorporated within the DDM framework. These models describe perceptual decision-making as a process in which noisy sensory evidence is gradually accumulated until a decision threshold is reached. By jointly modeling choices and response times, DDM-based approaches make it possible to estimate latent cognitive processes related to evidence accumulation.

The main objective of these analyses was to determine which computational mechanism better explained the confirmation bias observed in participants. In particular, models in which previous decisions biased the starting point of evidence accumulation were compared with models in which earlier choices directly altered the drift rate, meaning the way new sensory evidence was integrated during the decision process.

Importantly, these mechanisms make different predictions about participants' response times and decision behavior, allowing them to be distinguished in ways that are not possible using behavioral analyses alone.

Before fitting the models to the real behavioral data, simulations and parameter recovery analyses were performed to evaluate whether the models could reliably recover the parameters used to generate synthetic data. After validating the modeling pipeline, the models were fitted to the real dataset using hierarchical Bayesian estimation.

Finally, model comparison was carried out using predictive metrics such as WAIC (Widely Applicable Information Criterion) and LOO (Leave-One-Out Cross-Validation), allowing the comparison of different computational explanations for the observed behavioral biases.

6 Results

6.1 Behavioral results

6.1.1 General performance and response times

As a first approach to participants' behavior, response times and overall accuracy were analyzed across the different experimental conditions. Response time did not vary significantly across repetitions or trial types. In addition, accuracy also remained relatively stable across conditions and repetitions, indicating that the adaptive difficulty procedure worked properly.

Psychometric analyses showed that participants' decisions were strongly guided by the sensory evidence provided by the decision variable (DV). However, this relationship remained relatively stable across repetitions, as no significant changes were observed in the psychometric curves (Figure 1).

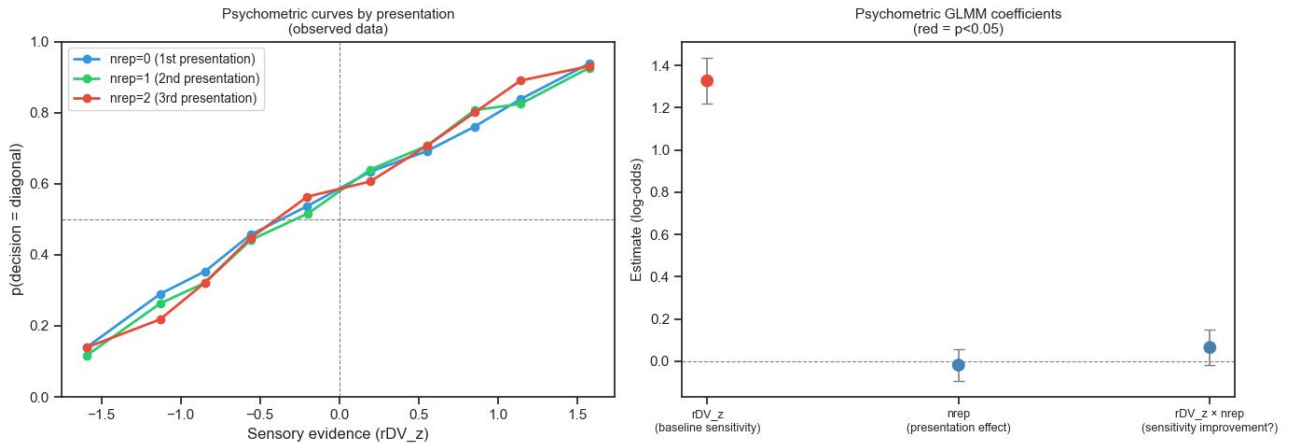


Figure 1: Psychometric curves across stimulus presentations (left panel) and GLMM fixed-effects coefficients for perceptual sensitivity (right panel). The left panel shows the probability of responding “diagonal” as a function of sensory evidence (rDV_z) for each presentation ($nrep = 0, 1, 2$).

6.1.2 Decision repetition and confirmation bias

The main goal of the behavioral analysis was to study how initial decisions influenced later perceptual responses. For this reason, the probability of repeating previous decisions across successive stimulus presentations was analyzed.

The results showed a strong tendency to maintain earlier decisions, with repetition probabilities reaching approximately 84–85% during the second and third presentations (Figure 2). This pattern appeared consistently across participants, indicating a high level of decisional stability throughout the task.

To determine whether this effect could simply be explained by perceptual memory or recognition of the exact same visual sequence, behavior was compared between repeat trials, in which the exact same sequence was presented again, and nonrepeat trials, in which a physically different sequence with similar sensory evidence was shown.

The analyses showed that participants continued to use current sensory evidence when making their decisions. However, repetition behavior remained highly similar in both conditions, suggesting that the effect was not only caused by seeing the same visual sequence again.

Although small differences across presentations were observed, the overall pattern remained stable: participants consistently tended to maintain their earlier decisions during later presentations. Similarly, the influence of sensory evidence remained comparable across repeat and nonrepeat trials.

On the whole, these findings support the idea that previous decisions bias later perceptual judgments even when the sensory sequence is not physically identical, consistent with confirmation bias mechanisms.

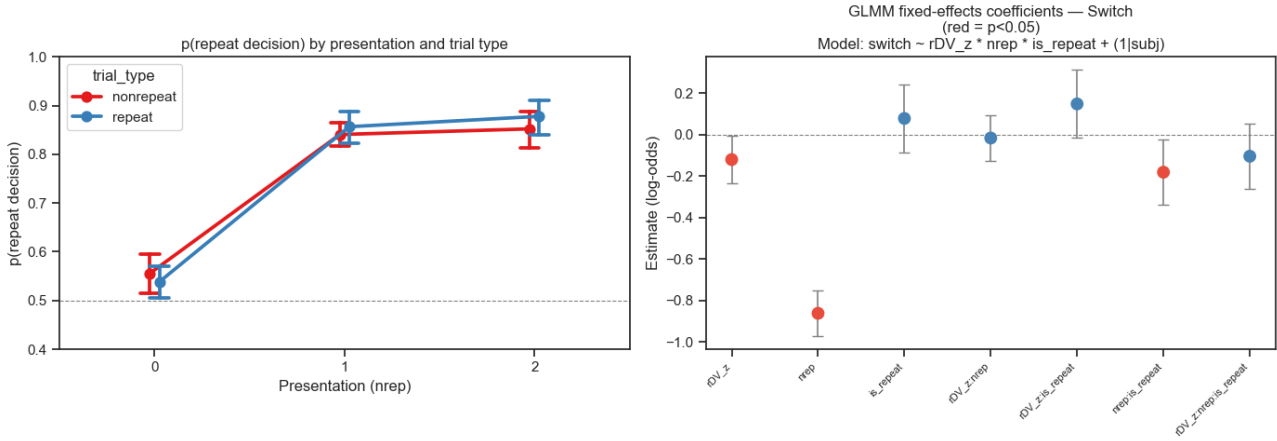


Figure 2: Probability of repeating previous decisions across repetitions for repeat and nonrepeat trials. The right panel shows the GLMM fixed-effects coefficients evaluating the influence of sensory evidence, repetition number and trial type on decision repetition behavior.

6.1.3 Confidence, metacognition and changes of mind

To study the role of metacognitive processes in perceptual decision-making, the relationships between subjective confidence, metacognitive efficiency, changes of mind and decision repetition behavior were analyzed.

The results showed that confidence in the previous decision strongly influenced whether participants maintained or changed their later response (Figure 3). In general, participants were more likely to repeat their previous decision when confidence was high, whereas changes of mind were associated with lower confidence levels. Similarly, the probability of repeating a decision increased progressively as confidence in the previous response increased, especially when the previous decision had been correct.

Exploratory analyses were also conducted to evaluate whether metacognitive efficiency was related to perceptual performance across repetitions. The relationship between meta- d' efficiency and accuracy varied across repetitions and depended on whether the previous response had been correct or incorrect (Figure 4).

When the previous response was correct, accuracy generally remained high across different levels of metacognitive efficiency. In contrast, when the previous response was incorrect, the relationship between metacognitive efficiency and accuracy showed more variable patterns across repetitions, with some conditions showing slight increases in performance and others showing the opposite tendency.

In summary, these effects were relatively weak and inconsistent, suggesting that metacognitive efficiency was not a strong predictor of decision repetition or changes of mind compared to subjective confidence.

In addition, changes of mind did not systematically improve perceptual accuracy, suggesting that changing a previous decision was not always beneficial under conditions of ambiguous sensory evidence.

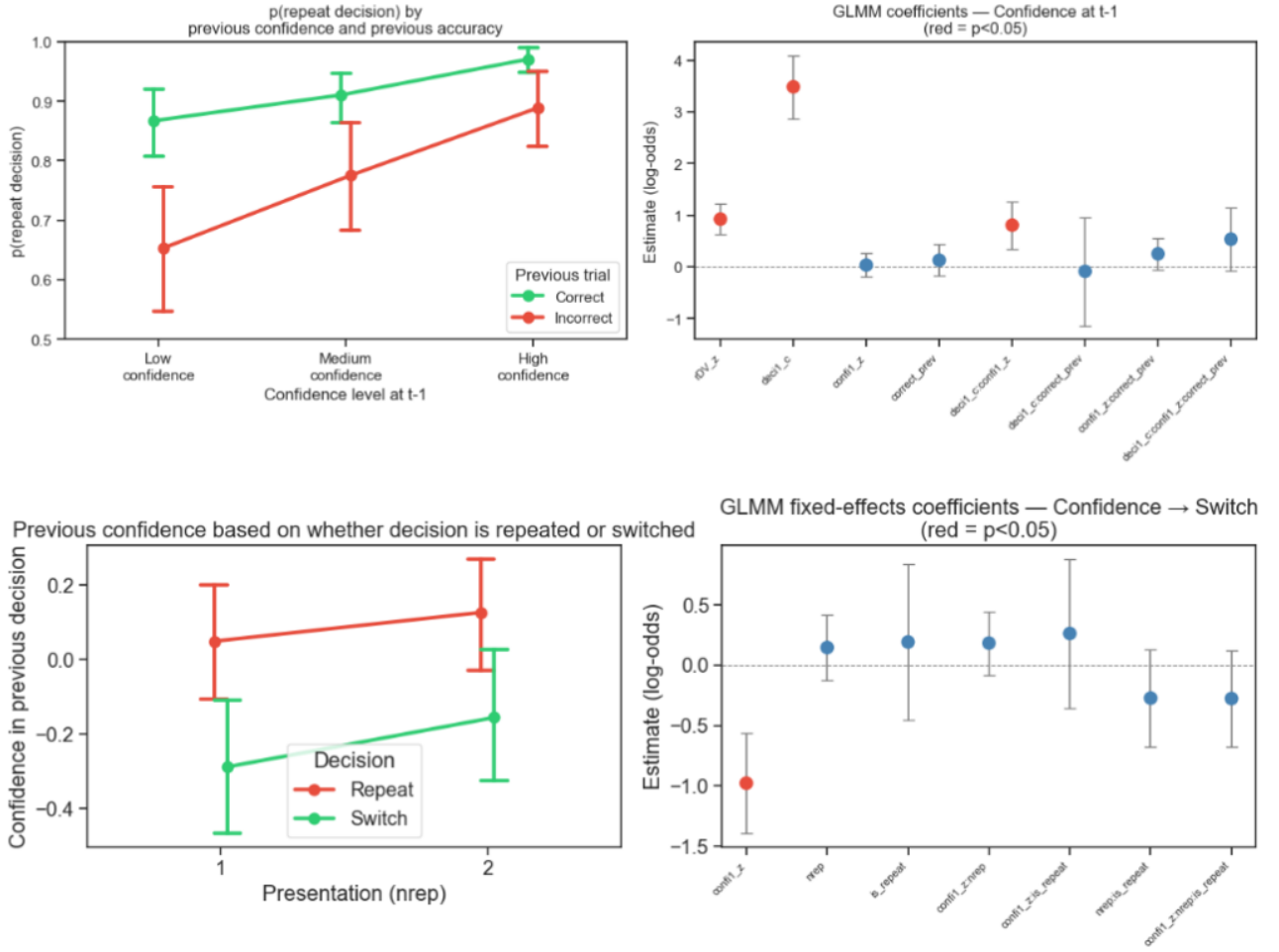


Figure 3: Relationship between confidence, previous accuracy and decision repetition behavior.

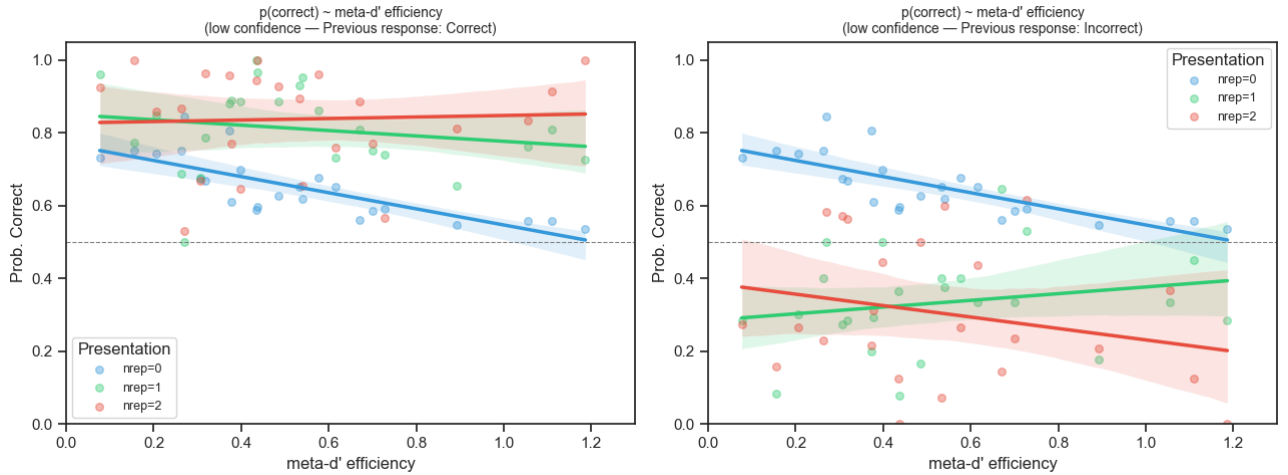


Figure 4: Exploratory relationship between metacognitive efficiency and perceptual accuracy across repetitions under low-confidence conditions.

6.1.4 Asymmetric processing of sensory evidence

To study whether participants were simply repeating previous decisions or whether previous decisions also influenced the processing of new sensory information, the contribution of each stimulus in the sequence was analyzed while controlling for the effect of the previous decision.

The results showed that all stimulus positions in the sequence significantly contributed to participants'

final responses, meaning that participants continued to use sensory information during the task. However, the influence of the previous decision was much stronger than the influence of any individual stimulus.

In particular, the effect of the previous response remained very large across repetitions ($\beta \approx 3$), whereas the contribution of individual stimuli was considerably smaller ($\beta \approx 0.3\text{--}0.4$) (Table 1). This suggests that participants were not relying only on current sensory evidence, but were strongly biased by what they had decided before.

These findings support the idea that previous perceptual decisions strongly shape later judgments, consistent with confirmation bias mechanisms.

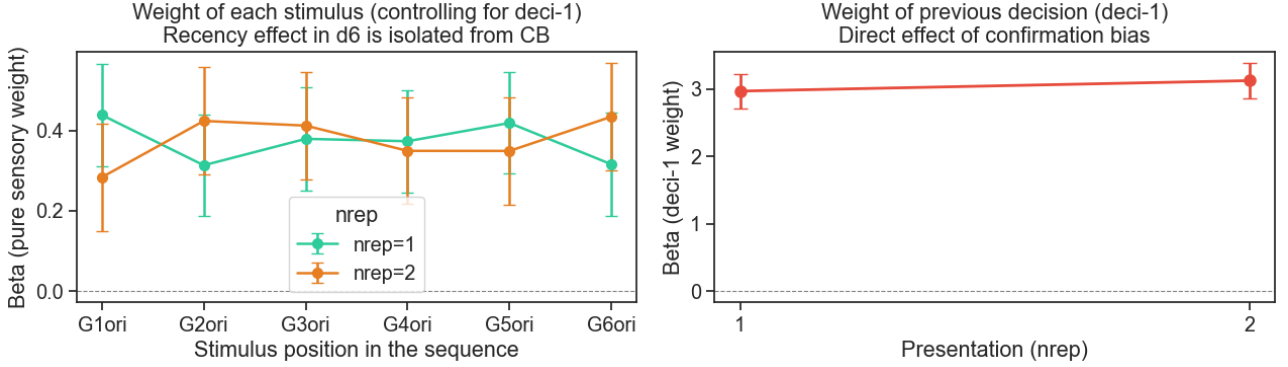


Figure 5: Contribution of sensory evidence and previous decisions to later perceptual responses.

Predictor	nrep = 1	nrep = 2
Previous decision	$\beta = 2.97^{***}$	$\beta = 3.13^{***}$
Individual stimuli	$\beta = 0.31\text{--}0.44^{***}$	$\beta = 0.28\text{--}0.43^{***}$

*** $p < .001$

Table 1: Relative contribution of previous decisions and sensory evidence.

6.1.5 Summary of behavioral results

In conclusion, the behavioral results showed a strong influence of previous responses on later perceptual judgments. Participants tended to maintain previous decisions across repetitions, while still integrating sensory evidence. However, the effect of the previous decision was much stronger than the contribution of individual sensory stimuli.

These findings motivated the later application of computational accumulation models to investigate the hidden mechanisms underlying the observed biases.

6.2 Results HSSM

6.2.1 Simulations and parameter recovery

Before applying the HSSM models to the real dataset, simulations and parameter recovery analyses were performed to evaluate whether the models could reliably recover known parameters introduced during data generation. The results showed satisfactory recovery of the main model parameters, particularly those related to drift rate and starting point, providing a methodological validation of the inference procedures used.

6.2.2 Fitting models to real behavioral data

After validating the models through simulations and parameter recovery, the HSSM models were fitted to the real behavioral data obtained during the perceptual task. The main goal of this analysis was to investigate which computational mechanisms could better explain the behavioral biases observed in participants.

On the one hand, models were tested where the previous decision modified the starting point of the decision process. On the other hand, models were evaluated in which previous responses altered the drift rate, suggesting that prior choices guide how new sensory evidence was accumulated during decision-making. The models were fitted using hierarchical Bayesian inference and included variables related to sensory evidence, initial decisions and subjective confidence.

6.2.3 Model comparison and computational interpretation

To determine which computational mechanism better explained the observed behavioral patterns, a formal comparison between HSSM models was performed using Bayesian model comparison metrics, including WAIC and LOO.

The results showed that models including drift-rate modulations provided a substantially better fit than models based only on starting point changes (Figure 6). Both WAIC and LOO strongly favored the drift-rate bias model, with differences greater than 200 points between models, indicating strong evidence in favor of the drift-rate mechanism (Table 2).

Importantly, the winning model showed that previous decisions had a strong influence on how new sensory evidence was accumulated during the decision process. In fact, the effect associated with previous choices was considerably larger than the effect of the current sensory evidence itself ($v_bias_cov = +0.765$ vs. $v_rdv_z = +0.361$), suggesting that participants tended to interpret new information in a way that supported their earlier decisions.

These findings indicate that the observed bias did not only reflect an initial response predisposition before evidence accumulation began, but also affected how sensory information was processed during the decision itself. From a computational perspective, this pattern is consistent with confirmation bias mechanisms in which earlier decisions bias the accumulation of later sensory evidence toward the previously selected interpretation.

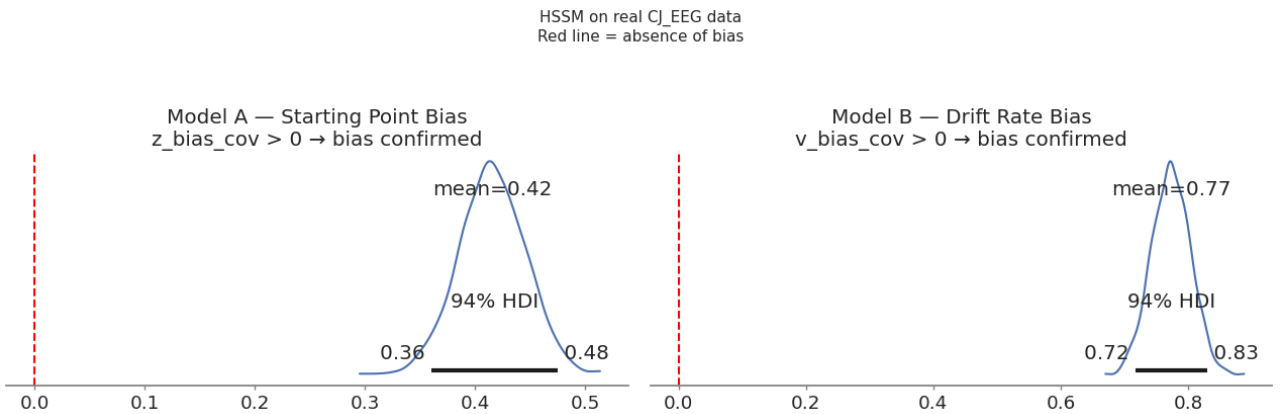


Figure 6: Comparison between the two computational bias mechanisms estimated with HSSM. Left panel: model in which previous decisions bias the starting point of the decision process. Right panel: model in which previous decisions bias the accumulation of new sensory evidence (drift rate). Positive values indicate confirmation bias effects. The black horizontal bars represent the estimated uncertainty intervals and the red dashed line indicates the absence of bias.

Model	WAIC	LOO
Starting Point Bias	-2614.48	-2614.49
Drift Rate Bias	-2376.14	-2376.16

Table 2: Bayesian model comparison between HSSM models. Higher (less negative) values indicate better predictive performance. Differences larger than 10 points are generally considered strong evidence in favor of one model.

6.2.4 Summary of computational results

The HSSM analyses complemented the behavioral findings and helped explain the computational processes underlying the observed bias. While the behavioral analyses showed a strong impact of prior decisions on later responses, the computational modeling helped explain how these effects emerged at the level of latent decision processes.

The simulations and parameter recovery analyses showed reliable recovery of the main model parameters, supporting the validity of the modeling approach. In addition, model comparison analyses suggested that the observed biases were better explained by drift rate modulations than by starting point changes alone.

Overall, these results support the idea that perceptual confirmation bias reflects an influence on how new sensory evidence is accumulated during decision-making rather than only a simple tendency to repeat previous responses.

Finally, the computational framework developed in this stage provides a useful basis for future machine learning analyses and for the later study of EEG correlates associated with perceptual decision-making.

7 Discussion

7.1 Confirmation bias and evidence accumulation

One of the main hypotheses of this project was that previous perceptual decisions would bias the integration of later sensory evidence during repeated perceptual judgments. More specifically, it was predicted that participants would show a tendency to maintain previous decisions even when new sensory evidence was presented, consistent with perceptual confirmation bias mechanisms.

The behavioral results strongly supported this prediction. Participants showed a high probability of repeating previous decisions across successive stimulus presentations and this effect remained present even when the exact visual sequence was not physically repeated. These findings suggest that the observed bias cannot be explained only by perceptual memory or simple stimulus repetition, but instead reflects a systematic influence of previous decisions on later perceptual processing.

A second prediction was that this bias would emerge mainly during the accumulation of sensory evidence rather than only as an initial decisional predisposition. Consistent with this hypothesis, the HSSM analyses showed that models including drift-rate modulations provided a substantially better fit to the data than models based only on starting point changes. In addition, the estimated parameters of the winning model revealed that the effect of previous decisions on the drift rate was considerably larger than the contribution of current sensory evidence.

From a computational perspective, these findings suggest that previous decisions influence how new sensory information is processed during the decision itself. Instead of evaluating the new evidence completely objectively, participants tended to interpret it in a way that supported their earlier choice. This interpretation is consistent with previous studies proposing that perceptual confirmation bias emerges because evidence accumulation becomes biased toward previously selected interpretations (Braun et al., 2018; Talluri et al., 2018; Urai et al., 2019).

7.2 Metacognition, confidence and decisional stability

The results suggest that metacognitive processes, particularly subjective confidence, play an important role in the stability and revision of perceptual decisions. Participants were more likely to maintain previous responses when confidence in the earlier decision was high, whereas changes of mind were more frequent when confidence was lower. This pattern supports the idea that confidence may act as an internal signal that influences whether people maintain or revise previous perceptual interpretations.

These findings are consistent with recent models proposing that confidence contributes to decisional stability by modulating how new sensory evidence is processed after an initial choice (Braun et al., 2018; Peters et al., 2017; Rollwage et al., 2020). In this context, high confidence may increase the tendency to interpret later evidence in a way that supports the previous decision, whereas lower confidence may facilitate changes of mind.

In contrast, exploratory analyses involving metacognitive efficiency did not show strong or consistent relationships with decision repetition behavior. Although some differences across repetitions and previous-response conditions were observed, these effects were relatively weak compared to the influence of subjective confidence. One possible explanation is that most participants showed very stable response patterns, making it more difficult to detect clear individual differences in metacognitive ability.

In addition, changing a previous decision did not consistently improve perceptual accuracy. This suggests that changes of mind were not always caused by participants detecting an earlier mistake, but also by other factors.

7.3 Limitations

Despite the consistency of the results obtained so far, the present work still has several limitations. Although the behavioral and computational analyses provide useful insights into perceptual confirmation bias, they do not directly identify the underlying neural processes, since no neural recordings have been analyzed at this stage of the project.

In addition, although HSSM models are useful for studying decision-making mechanisms, they represent simplified versions of perceptual processes. For this reason, the results obtained from these models should be interpreted cautiously and not as a direct description of how the brain exactly processes sensory information.

8 Current project status and future work

At the current stage of the project, the preprocessing and initial behavioral analyses of the data have been completed, together with the implementation of mixed-models and hierarchical evidence accumulation models (HSSM). The analyses carried out so far have identified consistent patterns of perceptual confirmation bias and provided an initial computational interpretation of the mechanisms involved in sensory evidence integration.

In addition, simulations and parameter recovery analyses have been performed to validate the computational models and compare different bias mechanisms related to starting point and drift rate modulations.

The next stages of the project will focus on developing machine learning approaches to predict participants' decisions from behavioral variables and computational model parameters. Later, EEG analyses will be incorporated to investigate the neural correlates associated with evidence accumulation and perceptual confirmation bias.

Finally, the last stages of the project will focus on integrating the different results, combining behavioral and neural analyses.

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A Computational analyses (HSSM)

A.1 Simulations and parameter recovery analyses

Before applying the HSSM models to the real behavioral data, simulations and parameter recovery analyses were performed to validate the computational pipeline and verify that the models could reliably recover the parameters used to generate the data.

Different versions of the DDM were implemented representing two possible confirmation bias mechanisms. In the first model, previous responses modified the starting point of the decision process, creating an initial bias before processing new sensory information. In the second model, previous decisions altered the drift rate, directly biasing the way new sensory evidence was integrated during decision-making tasks.

Using these models, simulated datasets were generated including perceptual decisions and response times similar to those observed in the real participants.

The simulations showed clear differences between both mechanisms.

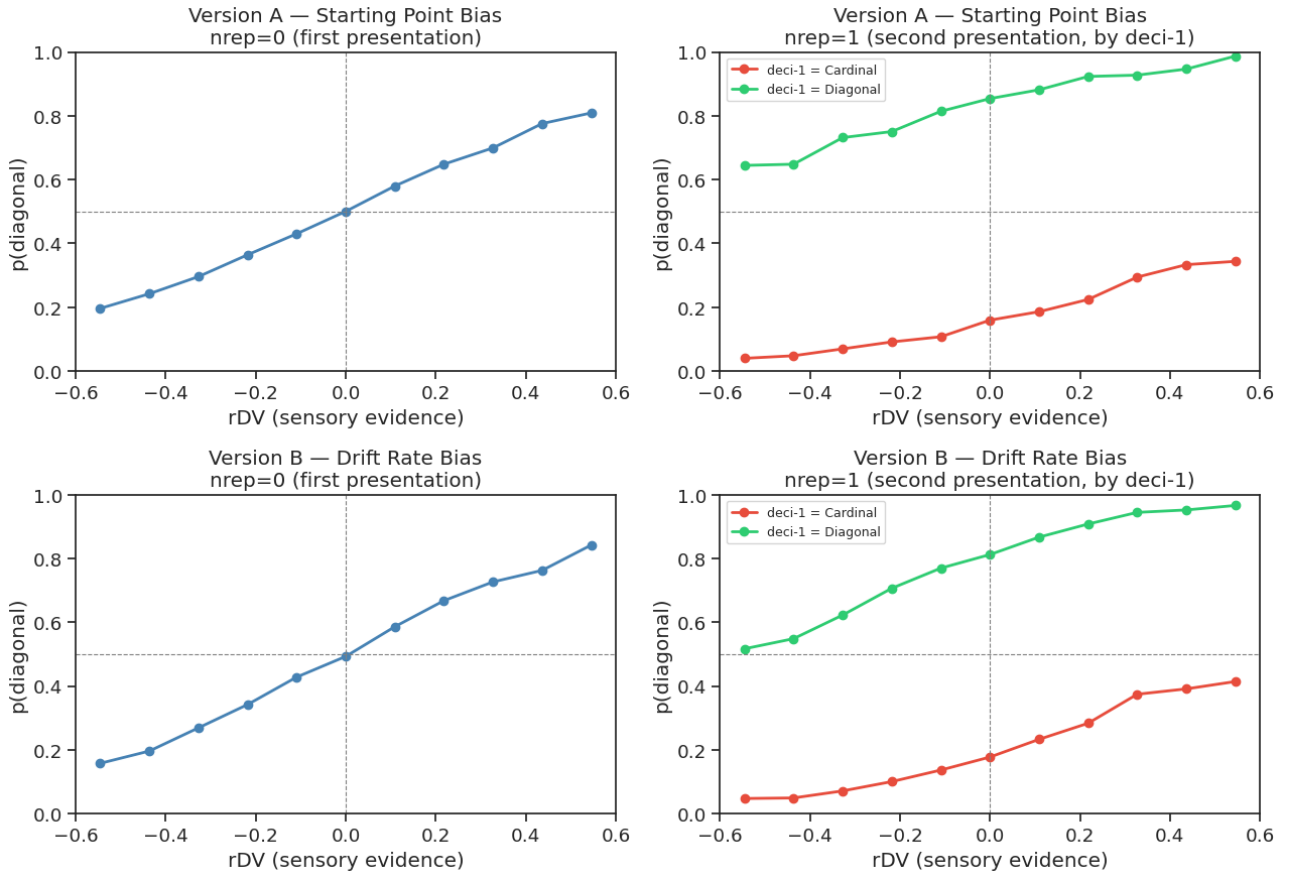


Figure 7: Simulated psychometric effects generated by starting point bias and drift-rate bias mechanisms within the DDM frameworks.

Models based on starting point bias mainly affected response times, while models including drift-rate bias produced stronger effects on perceptual behavior.

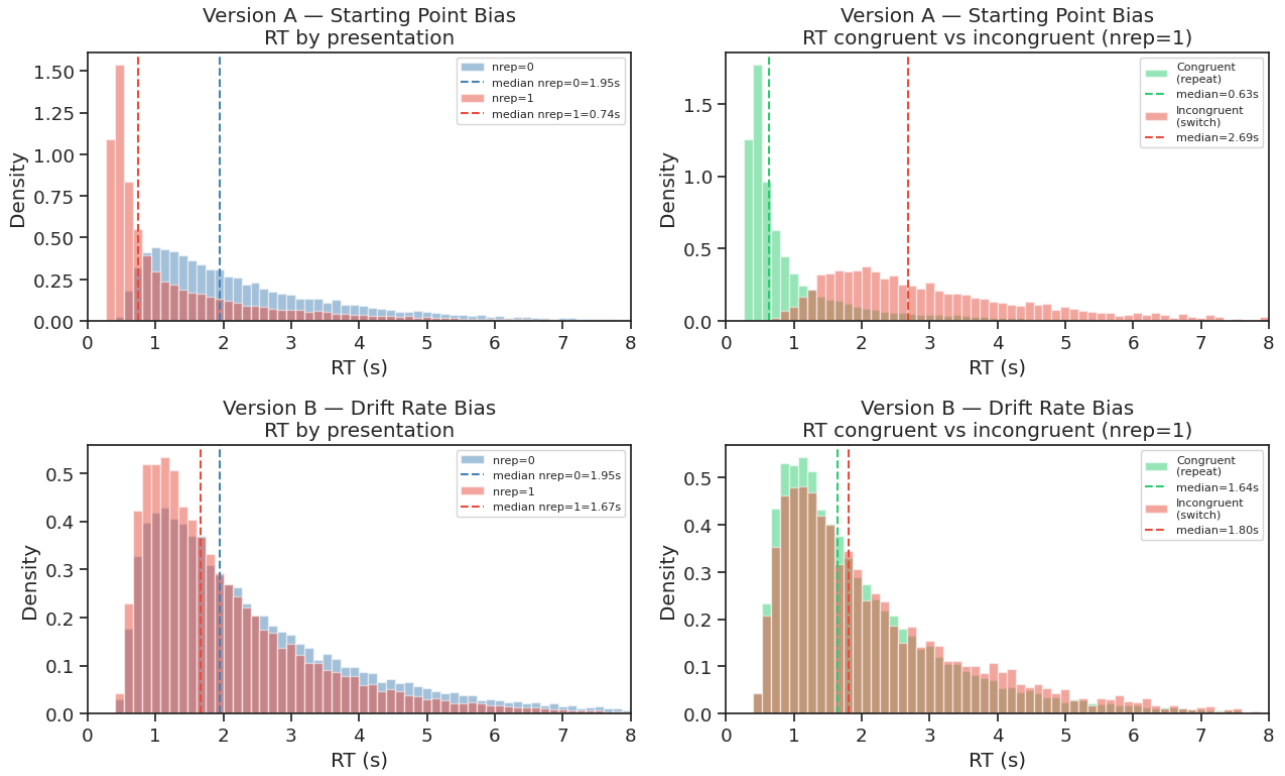


Figure 8: Simulated response time distributions for starting point bias and drift-rate bias models. Starting point bias produced stronger effects on response times, particularly between congruent and incongruent decisions.

Afterwards, the HSSM models were fitted again to the simulated datasets in order to evaluate whether the original parameters could be successfully recovered. The analyses showed good recovery of the main model parameters, indicating that the inference procedure was stable and reliable. Overall, this stage allowed the computational pipeline to be validated before fitting the models to the real behavioral data.

A.2 HSSM model comparison

After validating the models through simulations, the HSSM framework was applied to the real behavioral data obtained during the perceptual task. The main goal of this analysis was to investigate which computational mechanism better explained the behavioral biases observed in participants. To do so, models where previous decisions influenced different parameters of the evidence accumulation process were compared.

On the one hand, models in which previous decisions modified the starting point of the decision process were evaluated. On the other hand, models where previous decisions altered the drift rate were also tested, suggesting that previous decisions bias the processing of new sensory information. The models were fitted using hierarchical Bayesian inference and included variables related to sensory evidence, previous decisions and subjective confidence.

Model comparison was performed using predictive metrics such as WAIC and LOO. The results showed that models including drift-rate modulations explained participants' behavior better than models based only on starting point changes.

Overall, these findings suggest that perceptual confirmation bias is not only related to an initial tendency to repeat previous decisions, but also to an active modification in the way new sensory information is processed during decision-making.

A.3 Additional psychometric analyses

To further explore how participants' decisions evolved across repetitions, additional psychometric analyses were performed using the behavioral data obtained during the task.

First, psychometric curves were computed for each presentation of the stimulus sequence. Trials were grouped according to the value of the decision variable (DV), which reflected how strongly the stimulus sequence supported the cardinal or diagonal category. For each DV level, the probability of responding "diagonal" was calculated. This analysis allowed us to visualize how perceptual decisions changed as a function of sensory evidence across repetitions.

Additional measures based on Signal Detection Theory (SDT) were then calculated separately for each participant and repetition. Specifically, d' was used to estimate perceptual sensitivity, that is, how well participants discriminated between cardinal and diagonal stimuli. At the same time, the response criterion (c) was calculated to evaluate possible response biases across repetitions. These analyses allowed us to examine whether repeated presentations mainly affected perceptual sensitivity or instead produced changes in decisional bias.

Next, psychometric curves for the later repetitions were separated according to the participant's previous response. In other words, responses were analyzed depending on whether the participant had previously chosen the cardinal or diagonal option. The goal of this analysis was to examine whether previous decisions biased the interpretation of new sensory evidence during later repetitions.

Finally, GLMMs were fitted separately for each repetition and previous-response condition in order to estimate the intercept and slope of the psychometric curves. In this context, the intercept reflects a general tendency to favor one response independently of sensory evidence, whereas the slope reflects how strongly responses changed as sensory evidence became clearer. This analysis made it possible to evaluate whether previous decisions mainly produced shifts in decisional bias or whether they also altered sensitivity to sensory evidence.

Overall, the results suggested that previous decisions strongly influenced later perceptual responses, mainly by shifting participants' response tendencies across repetitions. In contrast, changes in perceptual sensitivity appeared smaller and less consistent across conditions.

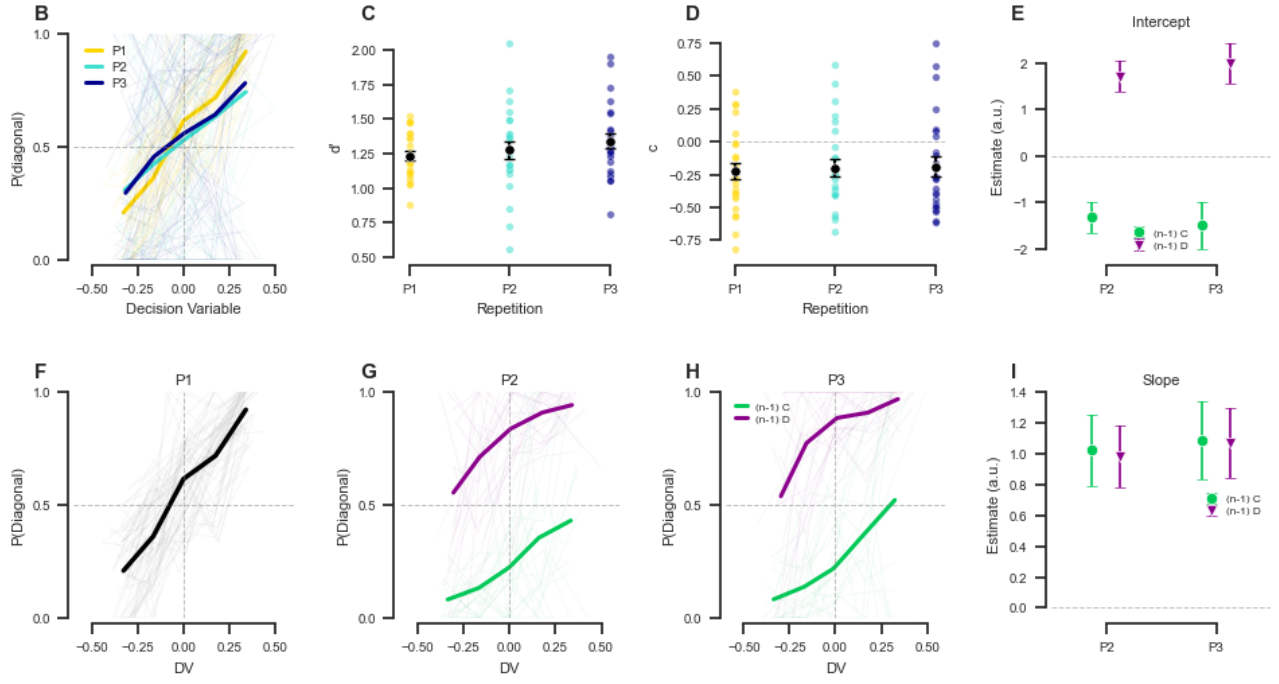


Figure 9: Additional psychometric analyses across stimulus repetitions. (B) Psychometric curves relating the decision variable (DV) to the probability of responding "diagonal" across repetitions (P1-P3). Thin lines represent individual participants and thick lines represent group averages. (C) Perceptual sensitivity (d') across repetitions. (D) Response criterion (c) across repetitions. (F-H) Psychometric curves separated according to the participant's previous response for each repetition. (E) GLMM intercept estimates showing how previous responses biased later decisions. (I) GLMM slope estimates showing how strongly current sensory evidence influenced responses.